

Instruct360: A 360° Video Framework for Egocentric Object Capture and Tracking for AR Instructions

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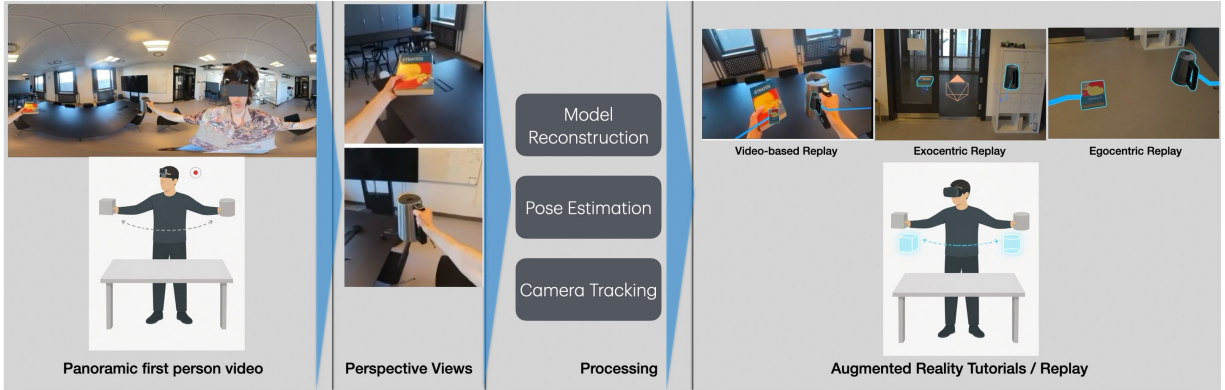


Figure 1: Egocentric 360 Object Capturing Framework: Left) 360 video input captured from an egocentric view using a head-mounted camera worn by the instructor. Center) Perspective views are extracted and processed using 3D model reconstruction, object pose estimation and camera pose tracking. Right) Extracted 3D objects are shown in AR tutorial visualization.

ABSTRACT

We present a framework for extracting 3D object representations and their spatial positions from egocentric equirectangular 360° video recordings. The framework uses only video captured by a head-worn 360° camera as input, without requiring additional sensors, depth measurements, or external tracking. It targets instructional scenarios in which object representation and positioning are central to learning. The extracted 3D objects are tracked over time and replayed in augmented reality (AR). By decoupling instructional information from the original camera viewpoint, learners can observe and follow procedures from their own perspectives. The primary motivation is to reduce constraints during capture, enabling natural head movement while preserving the spatial fidelity required for AR-based instructions.

Keywords: Augmented Reality, Instructions, Egocentric Capture, 360 Videos.

1 INTRODUCTION

Augmented reality (AR) has emerged as a promising interface for instructional and procedural learning [8, 9]. This is particularly the case in domains where spatial relationships and object interactions are critical to task understanding. A central challenge in authoring such AR instructional content lies in the capture process: conventional approaches often require carefully controlled camera viewpoints or constrained user motion to preserve spatial consistency, which can interfere with natural task execution and limit scalability [8]. Manually authored content comes with a long preparation time and requires modeling skills.

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Egocentric recording offers a compelling alternative by allowing instructors to perform tasks freely while wearing body- or head-mounted cameras [4], but translating such recordings into reusable, spatially grounded instructional representations remains an open problem. In this context, panoramic video provides a means to robustly observe the surrounding workspace despite head movement, while recent advances in object extraction and tracking enable the recovery of object-centric scene descriptions from video data [6, 7]. Building on these developments, in this work, we investigate how egocentrically captured panoramic video can be transformed into structured 3D object representations that support viewpoint-independent replay. By decoupling instructional content from the original camera trajectory, our framework aims to preserve spatial fidelity while enabling flexible, learner-centered visualization in AR. Specifically, we present Instruct360, a framework that explores the feasibility of extracting and replaying object-centered instructional information about everyday objects from egocentrically captured 360° video for augmented reality.

2 INSTRUCT360 FRAMEWORK

In order to explore whether instructional content can be extracted from head-worn 360° videos and used for AR replay, we developed a framework that reconstructs object 3D meshes and their poses at a given point in time from a single equirectangular 360° video stream captured by a head-worn camera. No additional depth measurements, camera poses, or other sensor data are required as input. The code of the framework is available online¹.

2.1 Object Capturing

The first step is to extract perspective views from the 360° panoramic video input frame and identify a list of object classes (Figure 2). The classes are not restricted to a predefined vocabulary (i.e., they can be specified as arbitrary semantic categories), allowing the user to declare which object types are relevant for the training interaction. Only these objects will be captured. Our framework

¹<https://github.com/ruetzmax/instruct360/>

thereby allows for the integration of visual models traditionally developed for planar images on panoramic inputs. A similar concept is used in both of the consecutive steps, mesh reconstruction (Figure 2, A) and pose estimation (Figure 2, B). Firstly, we determine relevant areas of the panoramic image frame by detecting the bounding boxes of the objects specified by the class input (Figure 2, A.1/B.1.1). This is done using SAM3 [2], a foundation model for promptable image segmentation. We determine the corresponding field of views (FOVs) and rotations on the image sphere of these areas, which allows us to reproject them to planar images (Figure 2, A.2/B.1.2). In addition to that, the camera poses are tracked using the ORB SLAM algorithm [3] following the implementation from OpenVSLAM [5]. The camera pose data will later be used for converting camera relative coordinates to world space (B.2).

2.2 Mesh Reconstruction

During mesh reconstruction, we only consider the first video frame containing an object of the respective input class. As described in the general framework, we extract planar images of relevant objects from the equirectangular input. Afterwards, boolean masks are generated using SAM3, which only contain the objects. Together with the planar RGB images, they are passed to SAM3D objects [6], a foundation model for reconstructing full 3D meshes from single input images. SAM3D outputs textured 3D object meshes, as well as poses for aligning the models relative to planar camera space (Figure 2, A.3). Since we know the rotations on the image plane for each planar image, we adjust the rotation of the planar camera-relative coordinates around the origin accordingly to get the omnidirectional camera-relative coordinates. These are now combined with the tracked world poses of the camera, which were obtained from ORB SLAM, to position the models in world space. This gives us a first global pose estimate for each reconstructed 3D object mesh.

2.3 Pose Estimation

Similarly to the process used in mesh reconstruction, we extract planar images for each object, only that this time, every frame in the input sequence has to be considered. Pose estimation is performed using the FoundationPose model [7], which can accurately perform 6DoF pose estimation on the basis of the mesh that was extracted from the first frame. Furthermore, we extract a boolean mask of the object using the SAM3 segmentation model. Lastly, we estimate a depth map of the image using Depth Pro [1], a model for zero-shot metric depth estimation, and we pass this to FoundationPose. We receive poses for each mesh in planar camera space (Figure 2, B.1.3), which we then transform to world space by the same method described in mesh reconstruction (Section 2.2).

2.4 Instruction Replay

Once the 3D meshes are reconstructed and their frame-to-frame pose estimation in world space is calculated, we replay of the captured object information in an AR application developed for the Meta Quest 2/3 headset. The reconstructed object meshes are displayed in 3D space, matching the poses that were extracted from the input video at the corresponding time. To better distinguish between real and virtual objects, all virtual objects are highlighted in a bright outline (Figure 1, Replay). In addition, we render a virtual trail to emphasize position changes over time. Passthrough of the HMDs camera feed is enabled to allow for incorporation of the information into the user’s current surroundings. AR Replay is implemented in one of three modes:

In **Exocentric Replay**, we display the 3D object information relative to a virtual placeholder representing the 360° camera that was used to record the input instructions. The placeholder mirrors the camera’s translation and rotation, while the reconstructed objects are placed accordingly in world space. This enables the user to

get an overview over the general course of motion from an outside perspective (Figure 1, Exocentric Replay).

In **Egocentric Replay**, no camera placeholder is displayed. Instead, we render the objects around the user, adjusting to the user’s own body position and orientation. This allows them to take the place of the person recording the input video, reenacting their motions and matching the real-world object poses with the virtual object poses (Figure 1, Egocentric Replay).

In the **Video-based Replay**, the egocentric replay is augmented with the input 360° video, which is rendered in the background instead of the camera passthrough. Suddenly, detail information that is not contained in the reconstructed objects can be incorporated into the training, such as complex hand-object interactions or the surrounding scene. The 360° video is stabilized using the previously tracked camera poses, creating a fixed frame of reference with the goal of reducing motion sickness (Figure 1, Video-based Replay). Switching between modes can be done during runtime, allowing for an intake of information at different levels of granularity.

3 RESULTS

We present preliminary results demonstrating the feasibility of recovering object-centered instructional information from two instructional sequences recorded in an office setting using a head-mounted 360° camera, featuring different objects and interaction scenarios (Figure 3).

Both sequences involve everyday object manipulation tasks with distinct object configurations, allowing us to explore the system under varying spatial layouts. The method produces temporally consistent 3D object representations that are replayed in an AR environment from novel viewpoints. Overall, the results indicate the feasibility of recovering and re-rendering object-centric instructional content from egocentric panoramic video.

4 LIMITATIONS

While FoundationPose supports full 6DoF pose estimation in principle, rotation estimation proves unreliable in practice. We attribute this to a combination of factors, such as occlusion by the user’s arm, loss of contextual reference when objects are cropped into planar image patches, mismatches between the reconstructed mesh and the actual appearance of the object in the image, and variations in illumination and shadowing conditions. These factors can influence the reliability of pose estimation and need to be considered when selecting the capturing conditions. As a result, all objects are fixed to the initial camera-relative rotation. Future work should investigate more robust pose estimation under varying lighting conditions and improved handling of appearance changes caused by shadows and illumination.

Furthermore, the initial 2D detection step is sensitive to the choice of class label. For some objects, detection fails with intuitive labels but succeeds when rephrased, e.g., when substituting ”iron” with ”white handheld object”. This suggests that these objects are underrepresented in the training data of the underlying detection model. They require manual label engineering, which reduces the out-of-the-box usability of the open-vocabulary approach. In addition, although the open-vocabulary formulation provides flexibility by allowing users to define relevant object categories, it requires an initial decision about which types of objects should be detected. Consequently, users need to have prior knowledge of the instructional content and relevant objects before capturing, representing a trade-off between flexibility in object selection and fully automatic discovery of relevant objects. Future work could investigate approaches that combine open-vocabulary detection with automatic discovery of relevant objects to reduce this dependency on prior knowledge.

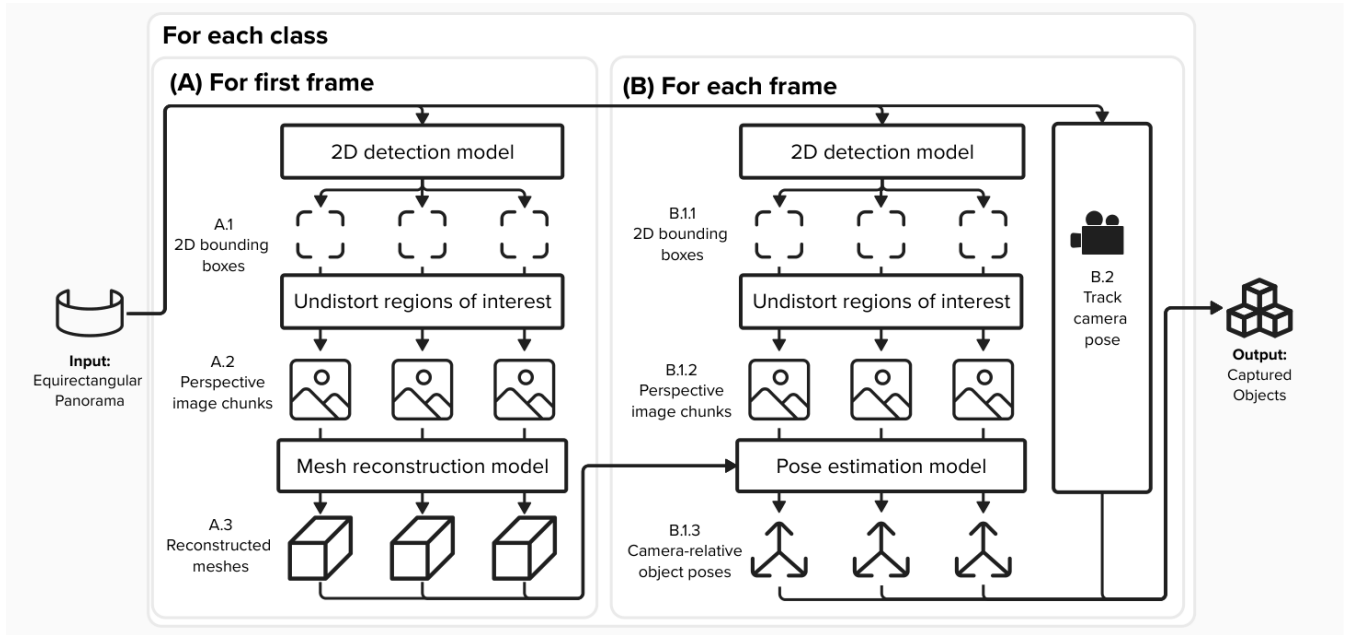


Figure 2: Object capturing framework for performing mesh reconstruction (A) and pose estimation (B). Relevant areas from the equirectangular input are extracted and undistorted to obtain perspective image chunks (A.1-2/B.1.1-B.1.2), to which conventional foundation models are applied. Reconstructed meshes (A.3) and camera-relative poses (B.1.3) are combined with the tracked camera pose (B.2) to compute the world-space captured objects.

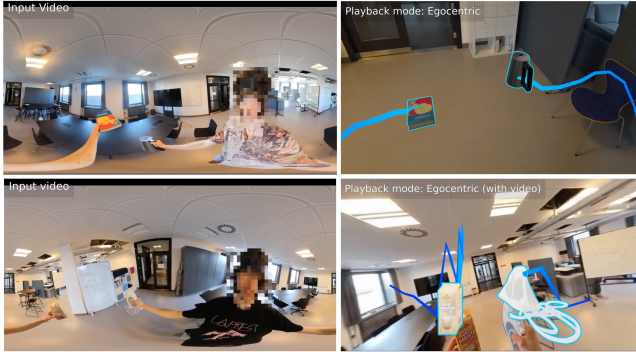


Figure 3: Preliminary results for two scenarios involving different everyday objects. Left) Input Sequences. Right) AR Replay for the corresponding input. Preliminary

5 CONCLUSION

In this work, we explore the feasibility of extracting and tracking object-centered 3D representations from egocentrically captured 360° panoramic video for use in AR-based instructional replay. By reconstructing objects and their spatial relationships independently of the original camera viewpoint, our Instruct360 framework enables instructional content to be revisited from flexible perspectives, supporting more natural capture conditions and potentially reducing the need for constrained head motion during recording. This decoupling of instruction from viewpoint is particularly valuable in scenarios where object configurations and manipulations are central to learning, such as physiotherapy with everyday objects.

At the same time, the presented results highlight several limitations that point to important directions for future work. Although the framework is designed to support full 6DoF pose estimation, reliable rotation estimation remains challenging in practice due to oc-

clusions, reduced contextual information in cropped object views, and discrepancies between reconstructed geometry and visual appearance. In addition, the sensitivity of the open-vocabulary detection stage to class label phrasing reveals limitations in current training data coverage and necessitates manual label engineering, reducing ease of use.

Overall, the approach provides a promising step toward object-centered spatial replay of instructional content in AR, while underscoring the need for more robust rotation estimation and improved open-vocabulary object detection. Addressing these challenges would strengthen the applicability of an egocentric capture framework for scalable and reusable AR instructional experiences.

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